

# 01

/ 13 LECTURES

## Introduction to Computer Networks

Why networks exist, how the Internet is built, and the vocabulary — delay, throughput, switching, devices & layers — that every later lecture builds on.

Why Networks

LAN · MAN · WAN

Internet Architecture

Edge vs Core

Packet vs Circuit Switching

Delay & Latency

Throughput & Bottlenecks

Hubs · Switches · Routers

OSI vs TCP/IP

### WHAT THIS LECTURE GIVES YOU

The complete ground floor of networking: what a network is made of, how the global Internet grew out of small local networks, how data moves (switching), how we measure "speed" (delay, latency, throughput), which devices do which job, and the two layer models used to talk about it all — with worked numericals and exam questions.

## LEARNING OBJECTIVES — AFTER THIS LECTURE YOU CAN

- Explain **why networks exist** and identify their building blocks (nodes & links)
- Classify networks by scale: **PAN, LAN, MAN, WAN** — with real examples
- Describe the Internet as a "**network of networks**" organised in ISP tiers
- Separate the **network edge** from the **network core** and name what lives in each
- Contrast **circuit switching** and **packet switching**, and argue why the Internet chose packets
- Compute **nodal delay** (processing, queueing, transmission, propagation) and **throughput** from formulas
- Pick the right device (**hub, switch, router, modem**) for a given job
- Read the **OSI (7-layer)** and **TCP/IP (4/5-layer)** models at a bird's-eye level

## Introduction to Computer Networks

### Why Networks?

Communication & resource sharing

Nodes + Links

Data exchange everywhere

### Network Types

PAN ~10 m

LAN — room / building

MAN — city

WAN — country / world

### Performance

4 delays + formulas

Latency & RTT

Throughput = min link

Bottleneck link

### Architecture

Edge: end devices

Access: your connection

Core: ISP mesh of routers

Tier-1 → Tier-2 → Tier-3

### Devices & Layers

Hub · Switch · Router · Modem

OSI 7 layers

TCP/IP 4 layers

### Switching

Circuit: dedicated path

Packet: independent hops

Store-and-forward

## 1 Why Computer Networks?

A **computer network** is a set of devices (called **nodes**) connected by communication channels (called **links**) that exchange data. That's the entire definition — everything else in this course is detail on top of it.

### Nodes — the devices

Computers, laptops, phones, servers, printers, and the networking boxes themselves: **routers, switches, hubs**. Any device that can send, receive, or forward data.

### Links — the connections

Wired (copper **twisted pair**, fiber optic) or wireless (Wi-Fi radio, cellular, satellite). A link carries signals that encode bits.

## Why do we need them?

- **Communication** — humans and programs talking across a room or across the planet (chat, email, calls, web).
- **Resource sharing** — one printer for an office, one file server for a team, one database for a million users.
- **Scale** — work distributed across machines: cloud computing, search engines, video streaming at planetary scale.
- **Cost & reliability** — shared infrastructure and replicated systems beat every user owning everything.

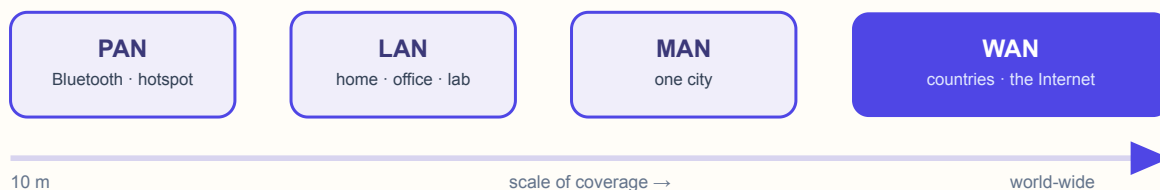
**ANALOGY** The classic mental model: the **postal system**. A letter (data) is placed in an envelope (packet) with an address (destination IP), handed to a post office (router), and forwarded office-by-office until it reaches the mailbox (destination device). No single postal worker knows the whole journey — each just knows the **next office**. This is exactly how Internet routing works (Lecture 7).

**BIG PICTURE** Everything you use daily — WhatsApp, UPI payments, YouTube, Instagram, exam results websites — is a program on one machine talking to programs on other machines through this stack of ideas. Master these 13 lectures and you can explain *every step* of what happens when you type a URL (Lecture 13 does exactly that).

## 2 Types of Networks — LAN, MAN, WAN

Networks are classified by **geographic scale**. The scale decides the technology, ownership, speed and cost.

| Type | Full Name                 | Coverage                    | Typical Owner            | Speed / Latency                                 | Examples                                                                |
|------|---------------------------|-----------------------------|--------------------------|-------------------------------------------------|-------------------------------------------------------------------------|
| PAN  | Personal Area Network     | ~10 metres                  | You                      | Fast, tiny range                                | Bluetooth earbuds, phone hotspot, smartwatch                            |
| LAN  | Local Area Network        | Room / building / campus    | One organisation         | High speed (100 Mbps–10 Gbps), very low latency | Home Wi-Fi network, computer lab, office network                        |
| MAN  | Metropolitan Area Network | A city                      | ISP / large org          | Moderate                                        | Cable TV network, city-wide ISP backbone, metro college campuses joined |
| WAN  | Wide Area Network         | Country / continent / world | Telecom companies (ISPs) | Varies; long distances → higher latency         | The Internet backbone, a bank's country-wide ATM network                |



**Figure 1.** Networks grow from personal to planetary — bigger scale means more owners, more hops, more latency.

**EXAM**

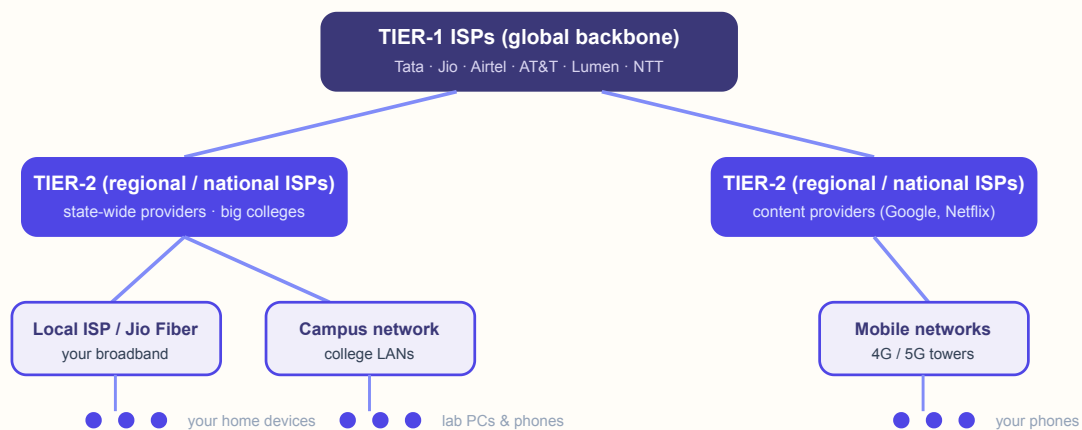
A favourite short-answer: *"Differentiate LAN and WAN."* Key points: LAN = small area, privately owned, high speed, low latency; WAN = huge area, owned by telecom providers (leased links), moderate speed, higher latency and cost. Give one example of each.

**NOTE**

**The Internet is not one WAN** — it is millions of LANs and WANs interconnected: a **network of networks**. A **VPN** (Virtual Private Network) is a secure tunnel built *on top of* the Internet to make a remote machine behave as if it is on your LAN.

### 3 Internet Architecture — the Network of Networks

No single company owns the Internet. It grew by connecting smaller networks (ISPs) into a loose global hierarchy:



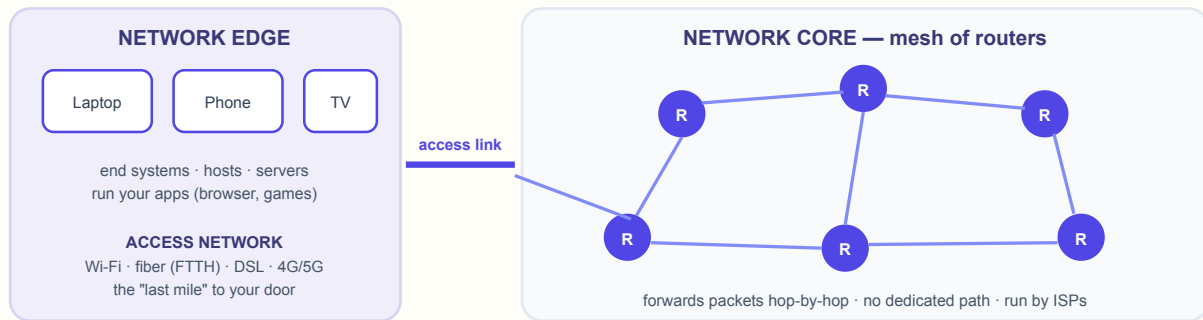
**Figure 2.** The Internet hierarchy: Tier-1 carriers form the core; regional and local ISPs bring access networks to your device.

- **Tier-1 ISPs** own long-distance fiber backbones and peer with each other as equals — they never pay anyone for transit.
- **Tier-2 / regional ISPs** buy connectivity from Tier-1s and sell to smaller networks.
- **Access networks (Tier-3)** — local ISPs, mobile towers, campus networks — connect the *last metres* to your device.
- **Content providers** (Google, Netflix, Cloudflare) run their own global networks and data centers, connecting into this hierarchy at many points (and via CDNs — Lecture 13).

**EXAM**

*"What is meant by the Internet being a network of networks?"* — Answer shape: end systems connect into access ISPs (LAN, then ISP); access ISPs connect into regional ISPs; regional into Tier-1 backbone ISPs; smaller networks at each level are themselves networks of routers and links. Nobody owns the whole; interoperability comes from shared protocols (TCP/IP).

## 4 Network Edge vs Network Core



**Figure 3.** Edge devices reach the core through an access link; the core is a cooperating mesh of routers running packet switching.

### Network Edge

End systems: PCs, phones, servers, IoT devices.  
**Hosts = clients** (requesters: your browser) **and servers** (responders: Google's machines). Edge devices run applications.

### Network Core

A mesh of interconnected **routers**. Its only job: **move data** — millions of packets per second, from any edge to any other edge. Contains no applications; pure forwarding machinery.

#### EXAM

"With a neat diagram, explain network edge and network core." Draw Figure 3. Edge = hosts + access networks (Wi-Fi, FTTH, DSL, cellular). Core = router mesh doing packet switching. One line each on client/server and on why the core is shared.

## 5 Circuit Switching vs Packet Switching

How does the core carry data for many conversations at once? Two historical answers:

### Circuit Switching — the telephone way

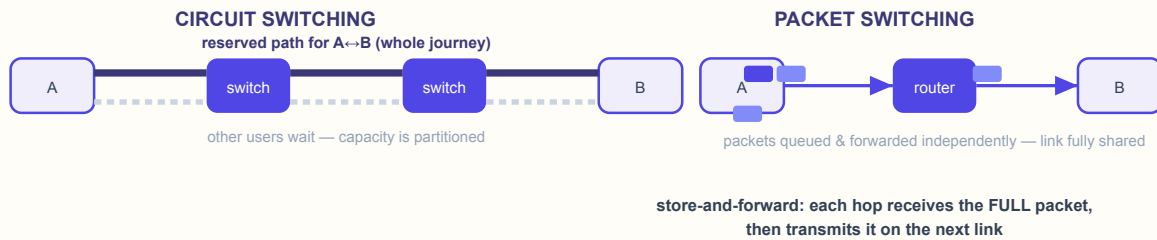
**Reserve a dedicated end-to-end path first** (signalling sets up the "circuit"), then send at fixed rate. Links are divided into channels by **FDM** (frequency-division: each call gets a frequency band) or **TDM** (time-division: each call gets repeating time slots).

- Guaranteed constant rate & latency — perfect for voice.
- Wasteful: silent moments (pauses in a call) still hold the circuit.
- Setup time before any data; fails if any intermediate trunk is busy.

### Packet Switching — the Internet way

**No reservation.** Data is chopped into **packets** (a few KB each) with headers; each router forwards them using **store-and-forward**: receive the whole packet, check it, send it on the next link. Packets from many users **statistically share** every link.

- Efficient sharing — idle capacity is instantly reused.
- Resilient: packets simply reroute around failures.
- No guarantees — congestion causes **queueing delay and loss**.



**Figure 4.** Circuit switching reserves a fixed channel end-to-end; packet switching forwards self-contained packets over shared links.

## Numerical: why packets win on efficiency

**Users share a 1 Mbps link. Each user is active 10% of the time and needs 100 kbps when active.**

**Circuit switching:** each user must reserve 100 kbps forever → only  $1 \text{ Mbps} \div 100 \text{ kbps} = 10$  users, and the link sits 90% idle for each of them.

**Packet switching (n users):** each user active with  $p = 0.1$ . Probability that more than 10 are active simultaneously:  $P(>10 \text{ active}) = \sum_{k=11..n} C(n, k) (0.1)^k (0.9)^{n-k}$ . For  $n = 35$ , this is under 0.04% — so packet switching supports **~35 users** on the same link with essentially no congestion.

**Total store-and-forward delay over N links =  $N \times (L / R)$**

$L$  = packet size in bits ·  $R$  = link rate in bps — each link must fully receive the packet before retransmitting it

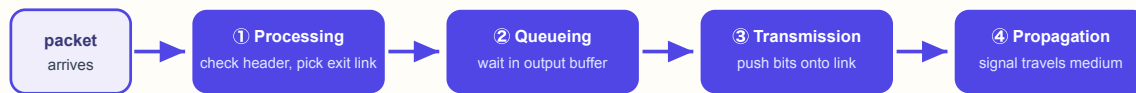
**NUMERICAL** **Classic exam numerical:** Send a 1,000-bit packet over two 1 Mbps links (store-and-forward). Total time =  $2 \times (1000 / 10^6) = 2 \text{ ms}$ . (First link: 1 ms to push it out; second router waits until it has all 1000 bits, then 1 ms more.) With a single 1 Mbps link it would be just 1 ms — each extra hop adds  $L/R$ .

### WHY THE INTERNET CHOSE PACKETS

Data traffic is **bursty** (long silences, sudden bursts). Reserving capacity for the silence wastes it; statistical sharing exploits it. Result: cheaper, scalable, resilient — at the cost of congestion management, which TCP handles later (Lecture 10).

## 6 Delay & Latency — the Four Delays

Every packet suffers delays at **every router hop**. Total **nodal delay** has exactly four components:



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

①+② happen in the router · ③+④ happen on the outgoing link

**Figure 5.** The delay pipeline every packet passes through at each hop.

| Delay               | What happens                                                     | Formula                    | Depends on                                                                    | Typical size                                    |
|---------------------|------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------|-------------------------------------------------|
| <b>Processing</b>   | Router examines header, checks bit errors, looks up output link  | $d_{\text{proc}}$          | Router hardware speed                                                         | Microseconds — usually ignored                  |
| <b>Queueing</b>     | Packet waits in the output buffer while earlier packets are sent | $d_{\text{queue}}$         | <b>Traffic intensity</b> (how busy the link is)                               | 0 when idle → grows unboundedly near congestion |
| <b>Transmission</b> | All $L$ bits of the packet are pushed onto the wire              | $d_{\text{trans}} = L / R$ | Packet length $L$ , link rate $R$                                             | ms scale for big packets on slow links          |
| <b>Propagation</b>  | The signal physically travels the medium's length                | $d_{\text{prop}} = d / s$  | Distance $d$ , propagation speed $s \approx 2 \times 10^8$ m/s (fiber/copper) | ~5 ms per 1000 km — physics, can't be upgraded! |

**TRAP**      **Transmission ≠ Propagation.** Transmission is writing bits onto the link (rate-limited by **bandwidth**); propagation is the signal racing along the wire (limited by **physics**). A 10-lane highway on-ramp analogy: transmission = how fast cars enter (one per second); propagation = how fast a car drives to the next city.

## Worked example — compute all four delays

**Given:** packet length  $L = 1000$  bytes, link rate  $R = 10$  Mbps, link length  $d = 1000$  km, propagation speed  $s = 2 \times 10^8$  m/s, queueing delay  $\approx 0$  (link idle), processing  $\approx 0$ .

- Convert units first**  
 $L = 1000 \text{ B} \times 8 = \mathbf{8000 \text{ bits}}$ ;  $R = 10 \text{ Mbps} = 10^7 \text{ bps}$ ;  $d = 1000 \text{ km} = 10^6 \text{ m}$
- Transmission delay**  
 $d_{\text{trans}} = L/R = 8000 / 10^7 = 8 \times 10^{-4} \text{ s} = \mathbf{0.8 \text{ ms}}$
- Propagation delay**  
 $d_{\text{prop}} = d/s = 10^6 / (2 \times 10^8) = 5 \times 10^{-3} \text{ s} = \mathbf{5 \text{ ms}}$
- Total nodal delay**  
 $d_{\text{nodal}} \approx 0 + 0 + 0.8 + 5 = \mathbf{5.8 \text{ ms}}$  — here propagation dominates

## Queueing delay & traffic intensity

$$\text{Traffic intensity } \rho = (a \cdot L) / R$$

$a$  = average packet arrival rate (packets/s) ·  $L$  = average packet size ·  $R$  = link rate

| Traffic intensity $\rho$ | What happens to average queueing delay                                                                   |
|--------------------------|----------------------------------------------------------------------------------------------------------|
| $\rho \ll 1$             | Arrivals are rare — packets rarely wait; average queueing delay $\approx 0$                              |
| $\rho \rightarrow 1$     | Delay grows rapidly and fluctuates wildly (bursts collide)                                               |
| $\rho > 1$               | Arrivals exceed service rate — queue grows without bound $\rightarrow$ <b>packets are dropped</b> (loss) |

**NUMERICAL Practice:**  $a = 1000$  packets/s,  $L = 1000$  bits,  $R = 2$  Mbps  $\rightarrow \rho = (1000 \times 1000) / (2 \times 10^6) = 0.5$ : stable, moderate queueing. If  $a$  doubles to 2000  $\rightarrow \rho = 1$ : queue explodes. Design rule: keep  $\rho$  well below 1.

## Latency vs RTT

- **Latency (one-way)** = time for a packet to travel source  $\rightarrow$  destination (all four delays summed over every hop).
- **RTT (Round-Trip Time)** = there and back. Ping reports RTT (Lecture 12). Bengaluru  $\rightarrow$  US server  $\approx 200$ –250 ms RTT.
- Latency matters as much as bandwidth for **interactive** apps: gaming, video calls, web page loads.

## 7 Bandwidth, Throughput & Bottlenecks

### Bandwidth

The **maximum** rate a link can carry (bits/second). What your ISP advertises: "100 Mbps plan". Also called link **capacity** or rate  $R$ .

### Throughput

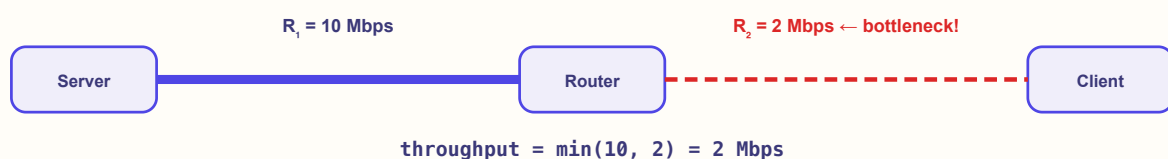
The **actual** rate achieved end-to-end at an instant: bits/second successfully delivered. What your download bar shows.

### Bottleneck link

The **slowest link on the path** — it alone caps the throughput. Fixing any other link changes nothing!

$$\text{End-to-end throughput} \approx \min(R_1, R_2, R_3, \dots, R_n)$$

the path is a chain of links — water can't flow faster than the narrowest pipe



**Figure 6.** Bottleneck principle — upgrading  $R_1$  (or your Wi-Fi) is pointless until the 2 Mbps link is fixed.

**ANALOGY** A **highway chain**: 8-lane expressway (high bandwidth) feeding a single-lane village road (low bandwidth). Traffic arrives at the village rate no matter how wide the expressway is. Same with your 1 Gbps laptop NIC on a 20 Mbps broadband line.

**NUMERICAL** **Path:** server– $R_1=30$  Mbps  $\rightarrow$  router – $R_2=10$  Mbps  $\rightarrow$  ISP – $R_3=50$  Mbps  $\rightarrow$  home. Throughput =  $\min(30,10,50) = 10$  Mbps. If ten clients share  $R_2$  simultaneously and fairly, each gets  $\approx 1$  Mbps. Also note: delay and throughput interact — a 1 MB file's transfer time is roughly  $\max(L/R_{\text{total}}, \text{RTT-limited window growth})$  — revisited in Lecture 10 (sliding window).

## 8 Network Devices — Hub, Switch, Router, Modem

### Hub

#### LAYER 1 · PHYSICAL

Dumb repeater: copies bits from one port to **every** other port. All devices share one collision domain and one bandwidth pool. Obsolete — but a favourite exam contrast.

### Switch

#### LAYER 2 · DATA LINK

Learns which **MAC address** lives on which port and forwards frames **only** to the right port. Each port gets full bandwidth; connects devices **within one LAN**.

### Router

#### LAYER 3 · NETWORK

Connects **different networks**; forwards packets using **IP addresses** and a routing table. The only device that lives in the network core. Your "home router" is router + switch + more (Lecture 11).

### Modem

#### EDGE · MODULATOR

**MOD**ulator-**DEM**odulator: converts between your home's digital signals and your ISP's line format (DSL phone line, cable TV coax, fiber light). The bridge onto the Internet.

| Question            | Hub                             | Switch                          | Router                                                     |
|---------------------|---------------------------------|---------------------------------|------------------------------------------------------------|
| OSI layer           | 1 — Physical                    | 2 — Data Link                   | 3 — Network                                                |
| Addressing used     | None — electrical repeat        | <b>MAC</b> (hardware) addresses | <b>IP</b> (logical) addresses                              |
| Forwarding decision | Floods to all ports             | MAC table lookup per port       | Longest-prefix match in routing table (L7)                 |
| Connects            | Devices in one segment (legacy) | Devices within one LAN          | <b>Different networks</b> (LAN $\leftrightarrow$ Internet) |
| Intelligence / cost | None · cheapest                 | Moderate                        | Highest — runs routing protocols                           |

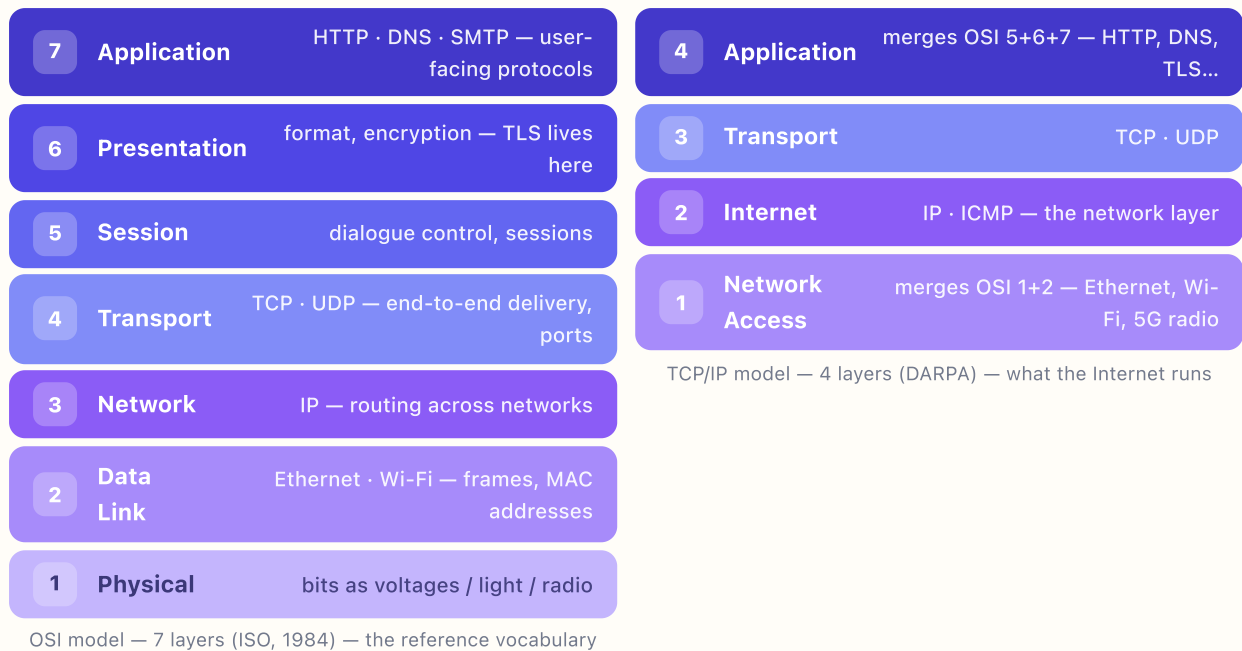
**ANALOGY** **Hub** = shouting in a room — everyone hears, only the addressee listens (chaos as crowd grows). **Switch** = an office receptionist who memorises faces (MACs) and hands each letter directly. **Router** = the postal system routing between cities (networks). **Modem** = the translator between your home's language and the ISP line's language.

## 9 OSI vs TCP/IP — a Bird's-Eye View

Complex systems need **layering**: each layer does one job, serves the layer above, and uses the layer below. Two models organise the ideas — **OSI** (7 layers, the vocabulary) and **TCP/IP** (4 layers, what the Internet actually runs). Full detail arrives in Lecture 2; today, meet the shape:

### Mnemonic — bottom to top

"**Please Do Not Throw Sausage Pizza Away**"  $\rightarrow$  Physical, Data Link, Network, Transport, Session, Presentation, Application. (Top-down: "All People Seem To Need Data Processing".)



**Figure 7.** Same ideas, different groupings: engineers implement TCP/IP but describe problems with OSI layer numbers ("that's a Layer 3 issue").

#### EXAM

"Compare OSI and TCP/IP models." Points: OSI = 7 layers, theoretical reference, created by ISO for standardisation; TCP/IP = 4 (or 5) layers, practical, developed by DARPA, what the Internet actually uses. OSI's Session+Presentation collapse into TCP/IP's Application; OSI's Physical+Data Link collapse into Network Access. We implement TCP/IP but **speak** in OSI layer numbers.

### LECTURE 1 — RAPID REVISION CARD

**Network** = nodes (devices) + links (wired/wireless) exchanging data.

**Internet** = network of networks: access ISPs → regional → Tier-1 backbone.

**Circuit** = reserved path (FDM/TDM), guaranteed but wasteful. **Packet** = store-and-forward sharing, efficient but queueing/loss.

**RTT** = round trip; ping measures it.  $\rho = aL/R < 1$  or queues explode.

**Hub L1** floods; **Switch L2** MACs in one LAN; **Router L3** IPs between networks; **Modem** bridges ISP line.

**PAN→LAN→MAN→WAN**: 10 m → building → city → world. LAN = fast, private; WAN = telecom-owned.

**Edge** = hosts/apps; **Access** = Wi-Fi/FTTH/4G; **Core** = router mesh, packet switching.

$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$ ;  $d_{\text{trans}} = L/R$ ;  $d_{\text{prop}} = d/s$  ( $s \approx 2 \times 10^8$  m/s).

**Throughput** =  $\min(R_1, \dots, R_n)$  — bottleneck link rules; bandwidth  $\neq$  throughput.

**OSI 7**: PDNTSPA · **TCP/IP 4**: App-Transport-Internet-Network Access.



## Important Questions — Exam Practice

### A. Multiple Choice

**Q1.** Which delay component depends on the physical distance between routers?

MCQ

- a) Processing
- b) Transmission
- c) Propagation
- d) Queueing

**Q2.** A 1500-byte packet is sent on a 3 Mbps link. Transmission delay is:

MCQ

- a) 0.5 ms
- b) 4 ms
- c) 2 ms
- d) 0.2 ms

**Q3.** Which device forwards frames using MAC addresses within a single LAN?

MCQ

- a) Hub
- b) Router
- c) Switch
- d) Modem

**Q4.** In packet switching, store-and-forward means the router:

MCQ

- a) sends bits as they arrive
- b) receives the entire packet before forwarding
- c) reserves bandwidth first
- d) broadcasts to all ports

**Q5.** A path crosses links of 8, 4 and 100 Mbps. End-to-end throughput is:

MCQ

- a) 100 Mbps
- b) 8 Mbps
- c) 4 Mbps
- d) 37.3 Mbps

### B. Short Answer (2–4 marks)

**Q6.** Define a computer network in terms of nodes and links. Give two reasons networks exist.

SHORT

**Q7.** Differentiate LAN and WAN (any four points) with one example each.

SHORT

**Q8.** Why is the Internet called a "network of networks"? Sketch the ISP tier hierarchy.

SHORT

**Q9.** Distinguish bandwidth, throughput and bottleneck link with an example.

SHORT

**Q10.** Differentiate transmission delay and propagation delay. Which one can an ISP upgrade?

SHORT

**Q11.** State the mnemonic for OSI layers and map the 7 OSI layers onto the 4 TCP/IP layers.

SHORT

### C. Long Answer / Numericals (5–8 marks)

**Q12.** Compare circuit switching and packet switching. Include FDM/TDM, efficiency, and a worked example showing why packet switching supports more bursty users.

LONG

**Q13.** A 1200-byte packet crosses three store-and-forward links:  $R_1 = 10$  Mbps (600 km),  $R_2 = 1$  Mbps (30 km),  $R_3 = 10$  Mbps (1200 km).  $s = 2 \times 10^8$  m/s. Compute per-link transmission and propagation delays, and the total one-way delay (ignore processing/queueing).

NUMERICAL

**Q14.** Users join a shared 10 Mbps link. Each needs 500 kbps when active and is active 20% of the time. (i) How many users can circuit switching support? (ii) Under packet switching, if 40 users subscribe, what is the probability that  $\geq 21$  are simultaneously active (comment qualitatively)?

NUMERICAL

**Q15.** "The network core is a mesh of routers running packet switching." With a diagram, explain edge vs access vs core, and explain what happens at one router as a packet transits it.

LONG

**Q16.** Explain the four components of nodal delay with formulas. Compute  $d_{\text{nodal}}$  for  $L = 8000$  bits,  $R = 100$  Mbps,  $d = 200$  km,  $s = 2 \times 10^8$  m/s, average queueing 3 ms, processing negligible.

LONG

#### ANSWER KEY

**A1.**  $c$  — propagation = distance  $\div$  speed of the medium.

**A2.**  $b$  —  $1500 \text{ B} = 12,000 \text{ bits} \div 3 \times 10^6 = 4 \text{ ms}$ .

**A3.**  $c$  — switch (L2, MAC table); hub floods, router uses IP.

**A4.**  $b$  — receive fully, then forward (adds  $L/R$  per hop).

**A5.**  $c$  — throughput = min link = 4 Mbps.

**A6.** Nodes = communicating devices; links = transmission media. Reasons: communication, resource sharing (any two: cost, reliability, scale).

**A7.** LAN: building/campus, privately owned, 100 Mbps–10 Gbps, very low latency (home Wi-Fi). WAN: country/world, telecom-owned leased links, higher latency & cost (Internet backbone, bank ATM network).

**A8.** Millions of interconnected networks; each access LAN  $\rightarrow$  local ISP  $\rightarrow$  regional  $\rightarrow$  Tier-1; no single owner; common protocols bind them. (Sketch as in Figure 2.)

**A9.** Bandwidth = capacity of a link; throughput = achieved end-to-end rate = min of links; bottleneck = the slowest link. Example: 100 Mbps Wi-Fi  $\rightarrow$  10 Mbps broadband = 10 Mbps.

**A10.** Transmission =  $L/R$  (push bits on, upgradeable by buying bandwidth); propagation =  $d/s$  (physics of distance, not upgradeable).

**A11.** Please Do Not Throw Sausage Pizza Away. 5+6+7  $\rightarrow$  Application; 4  $\rightarrow$  Transport; 3  $\rightarrow$  Internet; 1+2  $\rightarrow$  Network Access.

**A13.**  $d_{\text{trans}}$ : 0.96 ms / 9.6 ms / 0.96 ms ( $L=9600 \text{ b}$ ).  $d_{\text{prop}}$ : 3 ms / 0.15 ms / 6 ms. Total =  $(0.96+9.6+0.96) + (3+0.15+6) = 20.67 \text{ ms}$  — middle slow link dominates both terms.

**A14.** (i)  $10 \text{ Mbps} \div 0.5 \text{ Mbps} = 20$  reserved users. (ii) With  $n = 40$ ,  $p = 0.2$ : mean active = 8,  $\sigma \approx 2.53$ ; needing  $\geq 21$  active is  $z \approx 4.7$  standard deviations above the mean — probability  $\sim 10^{-6}$ , effectively never congested: all 40 share comfortably.

**A16.**  $d_{\text{trans}} = 8000/10^8 = 0.08 \text{ ms}$ ;  $d_{\text{prop}} = 2 \times 10^5/2 \times 10^8 = 1 \text{ ms}$ ;  $d_{\text{nodal}} \approx 0 + 3 + 0.08 + 1 = 4.08 \text{ ms}$ .

**A12, A15.** Structure your answer with Figures 4 and 3: definitions  $\rightarrow$  diagram  $\rightarrow$  3–4 contrast points  $\rightarrow$  example/numeric. Graders award labelled diagrams.